

COMPX551-26B Mobile and Wearable Computing

Assignment Four: Advanced Mobile and Wearable Development

Released: Monday 14th September | Due: Sunday 4th October 11:59pm

This is a group assignment, completed in teams of four. For this assignment, you are encouraged to use Kotlin, Jetpack Compose, Jetpack Compose for Wear OS, and Gemini inside Android Studio, along with the vendor SDK relevant to your team's chosen device (see below). As such, this assignment will be assessed via an **in-person (or live) code review and an in-class presentation**. You are required to submit your materials in advance and demonstrate your app running on real hardware during the review session. Please see the end of this assignment specification for further details.

Please do not enter the assignment specifications directly into Gemini (or any other Generative AI). The AI prompts should be self-written.

This assignment brings together everything from Assignments One to Three, this time using real hardware rather than emulators. Working in a team of four, you will build a comprehensive Android application that connects to one of three real devices: a Samsung Galaxy Watch, a Polar H10 heart rate sensor, or a Muse 2 EEG headband, and demonstrates substantial data processing, visualisation, and historical/summary reporting. Each team will be issued one Android phone and will choose one device to work with. Galaxy Watch teams must build a companion Wear OS app; Polar H10 and Muse 2 teams must integrate the relevant vendor SDK directly into the phone app.

A note on grading: this is the most substantial assignment in the course, and a working app is the floor, not the ceiling. Meeting each requirement functionally is not sufficient, on its own, for a high grade. To be considered for an A or A+, your application needs **real depth and polish across the whole pipeline: sophisticated, well-justified **data processing** (not just raw values passed straight through), a **visualisation** and **historical/summary view** that genuinely helps someone understand the data, and an overall standard of craftsmanship, complexity, and finish befitting a four-person team working on the final and most substantial assignment of an advanced, 500-level course. A simple, minimal application that technically satisfies every requirement will be graded as adequate at best; we are looking for real substance, creativity, and depth of understanding.**

Working as a Team

You will complete this assignment in a team of four. Team formation will be organised separately via Moodle.

To access your team's chosen device (Samsung Galaxy Watch, Polar H10, or Muse 2), one member of your group needs to go to CMS reception (G.1.21) to loan the equipment. Our admin team will ask for that member's name, student ID, and your group name.

Device numbers are limited (10 Galaxy Watches, 10 Polar H10s, and 5 Muse 2 headbands), and devices are allocated on a first-come, first-served basis. We recommend deciding on your team's preferred device early and loaning it out as soon as possible, and having a backup device in mind in case your first choice is no longer available.

Although your team will submit one application together, you will each be assessed individually as well as collectively:

- During the code review, each team member will be asked at least one **individual question** to demonstrate their own understanding of the whole application, not just the part they personally wrote.
- Your team must submit a **separate Contribution Statement** outlining what each member worked on (see Submission, below). Where contributions are significantly uneven, individual grades may be adjusted up or down to reflect this.

Project Description

Working in your team of four, you will build a comprehensive Android application for your team's chosen device. If your team is working with the Galaxy Watch, your application should include a companion Wear OS module. Rather than the strict paired-app architecture from Assignment Three, we'd prefer a hybrid approach: the watch app should be able to work on its own (standalone), but should offer a richer experience when paired with the Android phone app. If your team is working with the Polar H10 or Muse 2, your application can be a single Android phone app that connects directly to the device over Bluetooth using the vendor SDK; no separate wearable app is required for these two devices. Whatever your architecture, your application should include a clear welcome/dashboard screen and a way to start and stop a data-capture session.

Connect your app to your team's chosen device and capture live data. Galaxy Watch teams should capture at least three data types on the watch, including at least one from SensorManager and at least one from the Health Services Measure Client (as in Assignments One and Two), and send this data to the phone using the Wear OS Data Layer API, as in Assignment Three. Polar H10 teams should integrate the Polar BLE SDK and stream both heart rate and accelerometer data, with the option of including other available metrics such as RR intervals or ECG. Muse 2 teams should integrate the Muse SDK and stream EEG data (and, if you choose, other available metrics such as accelerometer or PPG data) directly to the phone app.

Polar H10 teams: use the official Polar BLE SDK, available at github.com/polarofficial/polar-ble-sdk, which includes the Android library, documentation, and example apps for connecting to Polar devices over Bluetooth.

Muse 2 teams: you will need access to the Muse SDK. Please contact Jemma directly to arrange this before you begin this part of the assignment.

Once your data reaches the phone, apply meaningful processing to it before it is displayed. Given the advanced nature of this course, we expect processing that goes beyond what was required in Assignment Three. Depending on your data type, this might include (not an exhaustive list): filtering or smoothing noisy signals; detecting events or thresholds (e.g. elevated heart rate, a distinctive EEG pattern, a motion event); and computing summary statistics such as min, max, average, or standard deviation over a session. Your processing should be a deliberate design decision that reflects genuine understanding of your chosen data type; be ready to explain and justify it during your code review.

Finally, display your processed data using a visualisation of your choosing. As in Assignment Three, this does not need to be a conventional chart; if a different visualisation suits your data better, or lets your team do something more interesting, feel free to use it. In addition to a live view, your application must also keep summary statistics or historical data across sessions (e.g. a log of past sessions with key summary statistics, or a longer running trend view); a single live-only view is not sufficient for this assignment. You may store this locally on the phone (e.g. using Room, a simple file, or SharedPreferences, depending on the scale of your data).

Please take screenshots (or short screen recordings) as you progress through development, and document the Gemini prompts you used along the way (see the "Assignment Four Assessment Information" section at the end of this document for more details).

In addition to building the application described above, your team will also deliver a short in-class presentation with slides; see "In-Class Presentation" in the Assessment Information section below for details.

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Assignment Four Assessment Information

Code Review:

This assignment will be assessed via an in-person (live) code review, worth 30% of your final course grade. This is a group code review: all team members must attend. If a team member does not attend, they will not receive a grade for the code review component. You are required to submit your materials in advance and demonstrate your app during the review session, with your full team present.

During your code review, you will be expected to:

Run and Demonstrate Your App

You will:

- Launch your application on your team's chosen device(s) in Android Studio
- Demonstrate its core features and behaviour, including live data capture, processing, visualisation, and historical/summary data
- Explain how the app satisfies the assignment requirements

Walk Through Selected Code

You will be asked to:

- Explain portions of your code chosen by the marker
- Describe how key components work (e.g., activities, composables, vendor SDK integration, Data Layer API usage, processing logic, visualisation, lifecycle handling)
- Justify design decisions your team has made, including your data type, processing approach, visualisation choice, and overall architecture

Individual Accountability

As this is a team assignment, you will also be assessed individually:

- Each team member will be asked at least one individual question to confirm their own understanding of the application as a whole, not only the part they personally worked on
- Team members who cannot account for parts of the application may have their individual grade adjusted accordingly, regardless of the team's overall result
- Your team's Contribution Statement (see Submission, below) will also be used to adjust individual grades where contributions were significantly uneven

Discuss Your Use of Generative Ai

You may be asked:

- How Generative AI supported your development process
- Why you accepted, modified, or rejected particular AI-generated suggestions
- How you ensured the correctness and quality of AI-assisted code

Respond to Improvement and Extension Questions

You will be asked one or more questions such as:

- How you might refactor or improve part of your code
- How the app could be extended with additional features (e.g. more data types, more advanced processing, or additional visualisations)

- How you would address performance, reliability, or battery-life issues

The code review will assess your understanding, not just whether the app works.

In-Class Presentation

In addition to the code review, your team will also deliver a short in-class presentation (including slides), worth 15% of your final course grade. Full requirements for the presentation will be provided in a separate specification.

Submission:

You are required to submit the following items for this assignment (one submission per team):

- **Application Submission:** submit the completed application (phone app, and Wear OS module if applicable). Please submit as a zip file.
- **Contribution Statement:** submit a short, separate statement (one per team) outlining what each member worked on during this assignment. This will be used, alongside the code review, to inform individual grade adjustments where contributions are significantly uneven.
- **AI Usage and Prompt Documentation:** you must also submit a brief supporting document (PDF or DOCX) that outlines how generative AI tools were used during development. Students are encouraged to use Gemini within Android Studio as a development aid. However, other Generative AI tools are also allowed.

The AI Usage and Prompt Documentation must include:

- Screenshots (or short recordings) of your app showing your progress at each key stage of development (setup, device connection, data processing, and final visualisation)
- A list of the prompts entered into AI that contributed meaningfully to the development of your app
- A short description of how the output was used (e.g., code generation, debugging, explanation of APIs, refactoring suggestions)
- Your data type, processing, and visualisation justifications described above

This document does not need to include AI's full responses verbatim, but the prompts should be sufficiently detailed to demonstrate how the tool informed your work.

Use of Generative AI and Academic Integrity:

Gemini is provided as a support tool and may be used to assist with learning, idea generation, and implementation. However:

- You are responsible for understanding, validating, and testing all submitted code
- The final submission must represent your own work and design decisions
- Blindly copying AI-generated code without comprehension is not acceptable

Failure to disclose the use of AI, or misrepresentation of AI-generated work as entirely human-authored, may be treated as a breach of academic integrity.

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